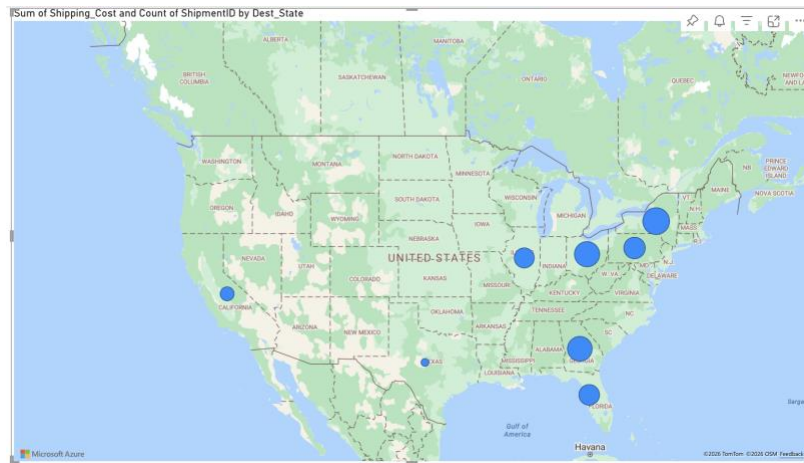


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- Assignment 5
- Assignment Open Date: March 4, 2026
- Assignment Due Date: March 18, 2026

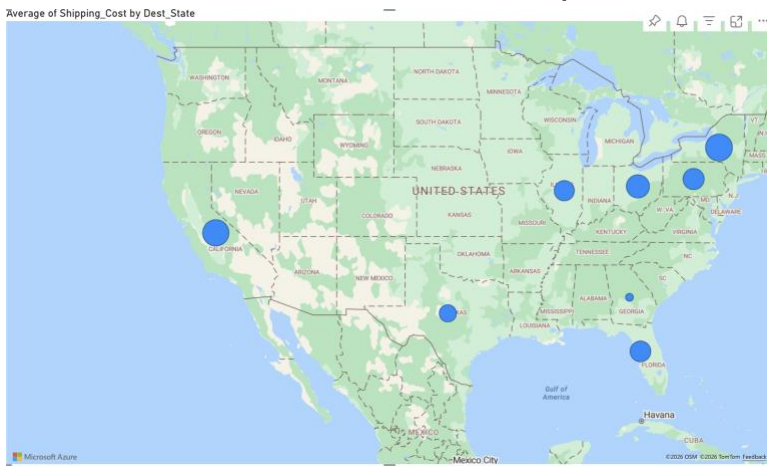
Visual 1: Shipment cost by destination



Managerial insight from the visualization

- We can see that shipping costs are not evenly distributed across states
- A few destination states account for the majority of logistics spending
- This indicates that demand is concentrated in specific geographic regions
- Higher costs in these areas likely result from longer shipping distances
- This suggests a potential mismatch between origin locations and demand regions
- Managers may consider adding regional distribution centers closer to high-cost states
- Focusing on these regions could significantly reduce overall shipping cost

Visual 2: Normalised (Average cost of shipping)



Managerial Insights (Normalized Map)

- This map shows average shipping cost rather than total cost, allowing fair comparison across destination states
- We now see which states are expensive per shipment, not just high volume
- California and northeastern states show higher average shipping cost, indicating potential inefficiencies
- Some states that looked expensive earlier may now appear moderate, meaning their costs were volume-driven
- This suggests that certain regions require cost optimization rather than demand management
- Managers should investigate routing, carrier selection, or distribution center placement for high-average-cost states

Why a map was appropriate for this analysis?

A map visualization was appropriate because the analysis focuses on geographic variation in shipping cost across destination states. Since supply chain performance is location-based, a map allows spatial patterns to be identified more clearly than tables or standard charts. The bubble sizes enable quick comparison between regions and highlight areas with higher logistics expenditure. This makes it easier to understand regional cost concentration and distribution patterns.

Difference between raw and normalized geographic views

The current visualization represents a raw geographic view, where bubble size reflects total shipping cost by destination state. Raw values highlight where overall spending is highest but may be influenced by shipment volume. Larger states with more shipments naturally appear more expensive. A normalized geographic view would adjust shipping cost relative to shipment volume, allowing fair comparison across states. This approach helps identify regions with higher cost per shipment rather than simply higher total cost.